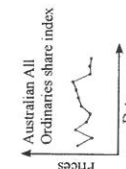
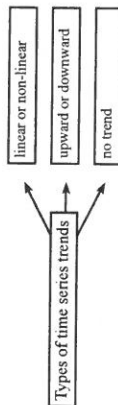


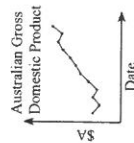
Time Series

A time series is a sequence of data measured at regular time intervals.

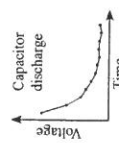
Time series are usually visually represented by data points joined by a line.



The above time series plot indicates no trend.



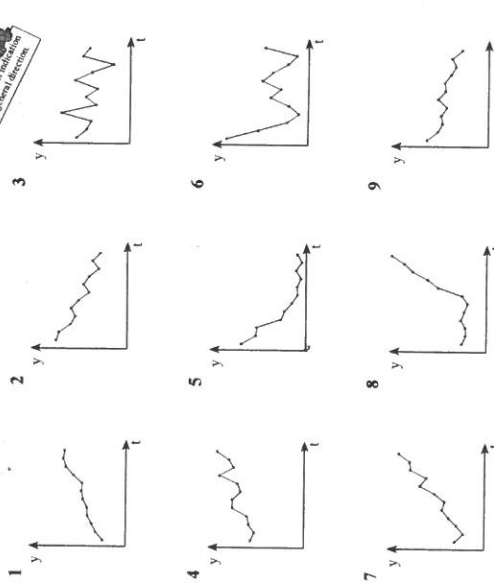
The above time series plot indicates a linear upward trend.



The above time series plot indicates a non-linear downward trend.

Exercise 18.4

Describe each of the following time series plots:



Exercise 18.5

1 Rainfall has a significant impact on many aspects of Australian life and there is a keen interest in weather forecasts.

- Draw a time series plot of the data.
- Describe the time series plot.
- Attempt to make a forecast of the mean annual rainfall for the next five years.
- How confident are you of your forecast?

Year	mm	Year	mm
1905	420	1960	450
1910	460	1965	420
1915	420	1970	420
1920	430	1975	650
1925	410	1980	470
1930	470	1985	450
1935	400	1990	420
1940	480	1995	400
1945	430	2000	700
1950	560	2005	430
1955	530	2010	700

2 Temperature also has a significant impact on many aspects of Australian life and there is a keen interest in weather forecasts.

- Draw a time series plot of the data.
- Describe the time series plot.
- Attempt to make a forecast of the mean annual temperature for the next five years.
- How confident are you of your forecast?

Year	°C	Year	°C
1905	21.6	1960	21.6
1910	21.6	1965	21.6
1915	21.5	1970	21.7
1920	21.5	1975	21.7
1925	21.4	1980	21.8
1930	21.5	1985	22.0
1935	21.6	1990	22.1
1940	21.5	1995	22.1
1945	21.4	2000	22.2
1950	21.5	2005	22.3
1955	21.6	2010	22.3

3 It is thought that sea surface temperatures affect the Australian climate. The following table refers to sea surface temperatures in the Australian region.

- Draw a time series plot of the data.
- Describe the time series plot.
- Attempt to make a forecast of the mean annual sea surface temperatures for the next five years.
- How confident are you of your forecast?

Year	°C	Year	°C
1905	21.4	1960	21.8
1910	21.5	1965	21.7
1915	21.5	1970	21.8
1920	21.5	1975	21.9
1925	21.5	1980	21.9
1930	21.5	1985	21.8
1935	21.6	1990	22.1
1940	21.7	1995	22.0
1945	21.7	2000	22.3
1950	21.6	2005	22.2
1955	21.6	2010	22.3